Assignment No-1

1) Write a Java program using the `InetAddress` class to display the IP address of a given host name.
2) Write a Java program using the `URL` class to read and display different components of a given URL.
3) Write a Java program using `Socket` and `ServerSocket` to implement one-way communication between a client and a server.
4) Write a Java program using the `java.net` package to implement a multi-client server that handles client requests using sockets and streams.
5) Write a program develop a simple **two-way chat application** using `Socket` and `ServerSocket` where:
   - The server waits for the client connection
   - Both client and server can send and receive messages until either side types `exit`

6) Write a Java program to connect to a database using JDBC and display all records from a table using a `Statement` and `ResultSet`.

7)Write a Java program using **JDBC** to insert, update, and delete records in a database table using `Statement` objects with proper exception handling.

8)Write a Java program using **PreparedStatement** to insert and retrieve records from a database and demonstrate **ResultSet navigation**.

9)Write a Java program using **JDBC** to connect to a database, execute CRUD operations using `PreparedStatement`, and call a stored procedure using `CallableStatement` with proper exception handling.

10) Write a Java program to establish a JDBC connection and display all records from a database table using `Statement` and `ResultSet`.

11) Write a Java program using JDBC to insert records into a database table using a `Statement` object.

12) Write a Java program using JDBC to update existing records in a database table using a `Statement`.

13) Write a Java program using JDBC to delete records from a database table using a `Statement`.

14) Write a Java program to demonstrate exception handling in JDBC while connecting to a database.

15) Write a Java program using `ResultSet` to navigate data using `next()`, `previous()`, and `first()` methods.

16) Write a Java program using `PreparedStatement` to insert records into a database securely.

17) Write a Java program using `PreparedStatement` to retrieve records based on user input.

18) Write a Java program to demonstrate the use of different types of JDBC drivers.

19) Write a Java program using `CallableStatement` to execute a stored procedure from Java.

Assignment No-2

1) Write a Java Servlet program to demonstrate the **Servlet life cycle** using `init()`, `service()`, and `destroy()` methods.

2) Write a Servlet program using **GenericServlet** to display a simple message in the browser.

3) Write a Java program using **HttpServlet** to handle HTTP GET and POST requests.

4) Write a Servlet program to demonstrate the use of the **RequestDispatcher** interface with `forward()` method.

5) Write a Servlet program to demonstrate the use of **RequestDispatcher** with `include()` method.

6) Write a Servlet program to implement **session tracking using Cookies**.

7) Write a Servlet program to implement **session tracking using Hidden Form Fields**.

8) Write a Servlet program to demonstrate **URL Rewriting** for session management.

9) Write a Servlet program using **HttpSession** to store and retrieve user information.

10) Write a Java Servlet program to demonstrate **session management** and explain the advantages of **HttpSession**.

11) Write a Servlet program to implement session management using **URL Rewriting**.

12) Write a Servlet program to store and retrieve user information using **HttpSession**.

13) Write a **Servlet program** that connects to a database using **JDBC** and displays all records from a table in the browser.

14) Write a **Servlet program** using **HttpServlet** and **PreparedStatement** to insert user input from a form into a database.

15) Write a **Servlet program** that retrieves data from a database and displays it using **ResultSet** in an HTML table.

16) Write a **Servlet program** to update database records using **JDBC** based on user input from a form.

17) Explain how **Servlets can interact with databases** using JDBC, and discuss the advantages of using Servlets over CGI for database-driven web applications.

Assignment No.3

1) Write a JSP page to display "Hello World" using **expression tag** `<%= %>`.

2) Write a JSP page to display the current date and time using a **scriptlet**.

3) Write a JSP page using **declarative tag** to declare a method that returns a greeting message.

4) Write a JSP page to demonstrate the use of **request and response implicit objects**.

5) Write a JSP page to demonstrate the use of **session and application objects**.

6) Write a JSP page to demonstrate **exception implicit object** handling by generating an exception.

7) Create a **JavaBean** with `name` and `age` properties and display them in a JSP using `<jsp:useBean>` and `<jsp:getProperty>`.

8) Write a JSP page to **set values in a JavaBean** using `<jsp:setProperty>` and display them.

9) Write a JSP page to demonstrate **JavaBean in a form submission** scenario.

10) Write a JSP page using **JSTL core `<c:if>` tag** to display a message if a number is positive.

11) Write a JSP page using **JSTL `<c:choose>` tag** to display "Pass" or "Fail" based on a marks value.

12) Write a JSP page using **JSTL `<c:forEach>` tag** to display elements of an array.

13)Write a JSP page using **JSTL `<c:set>` and `<c:out>` tags** to store and display a value.

14) Write a JSP page using **JSTL formatting `<fmt:formatNumber>` tag** to display a number in currency format.

15) Write a JSP page using **JSTL `<fmt:formatDate>` tag** to display the current date in `dd/MM/yyyy` format.

16) Write a JSP page using **JSTL XML tags** to parse and display data from a small XML file.

17) Write a JSP page using **JSTL SQL `<sql:query>` tag** to fetch all records from a database table and display them in an HTML table.

18) Write a JSP page using **JSTL SQL `<sql:update>` tag** to insert a record into a database table.

19) Write a JSP page using **JSTL SQL `<sql:param>` tag** to safely pass a parameter to a query.

Assignment No-4

1) Write a Hibernate program to **connect to a database and retrieve all records** from a table using **annotations**.

2) Write a Hibernate program to **insert a record** into a database using **annotations**.

3) Write a Hibernate program to **update a record** in a database table using Hibernate session.

4) Write a Hibernate program to **delete a record** from a table using Hibernate session.

5) Write a Spring program demonstrating **constructor-based DI**.

6) Write a Spring program demonstrating **setter-based DI**.

Assignment No-5

1) To develop a **web-based application** using **JDBC, Servlet, and JSP (MVC architecture)** that allows users to **add, update, delete, and view student records** efficiently.

- Implement **MVC architecture**:
  - **Model:** JDBC classes to interact with the database
  - **View:** JSP pages to display and collect data
  - **Controller:** Servlet classes to handle requests and responses
- Perform **CRUD operations** on the Student table.
- Demonstrate **form validation, session management, and dynamic data display**.
- Use **RequestDispatcher** to forward requests to JSP views.

2. Student Management System (SMS)

Description:
Create a **web application** to manage student records using **Servlet + JSP + Hibernate + Spring**.

- Features: Add, Update, Delete, View students
- Use **MVC architecture**: Servlet → Controller, JSP → View, Hibernate → Model, Spring → Service Layer
- Optional: Search student by ID or name, login/logout functionality.

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3. Employee Management System (EMS)

Description:
Develop a **web-based employee management application**.

- Features: Add, Edit, Delete, View employee records, search by department
- Technology: Servlet + JSP + Hibernate + Spring
- Implement **service layer using Spring** and ORM using Hibernate.

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4. Online Library Management System

Description:
Create an application to manage library books and borrowing details.

- Features: Add books, issue books to students/employees, return books, display all books
- Technologies: Servlet + JSP + Hibernate + Spring
- Use **MVC**, Hibernate for DB operations, and Spring for service/transaction management.